

ADA-Marker Answer Sheet

Anonymized local demo

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PROBLEM

Q3

Answer

7. let's use DP to record everything

Let $D(i, j)$ be the path length from vertex i to vertex j

our goal is to choose $\max_{i \leq j} D(i, j)$ $1 \leq i \leq j, i, j \leq n$, we can draw a chart

like this

	1	...	n
1	0		
...			
i	x	0	0
...			
n			0

and x means there's no path from

i to j

we m is $n-1$ so $O(m+n) = O(2n) = O(n)$

so if we full the chart we just need $O(n)$ times because

we can use the record we pre-recorded. Since it's acyclic so $D(1, 1) = 0$

Finally we just pick the biggest table, the worst case

is $2, 3, 4, \dots, n$ then we can't use pre recorded record but it's still

$O(n)$ operations and do ~~the~~ problem.